

Linear Programming Report

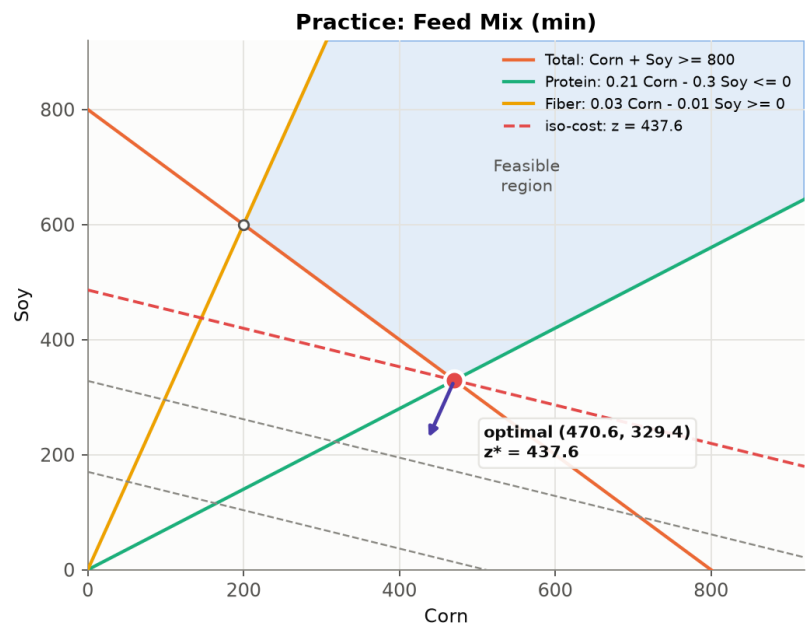
Sensitivity Analysis & Duality

Practice: Feed Mix (min)

A farmer blends corn (x1) and soybean meal (x2), in kg, to meet protein and fiber requirements at minimum cost. Solve graphically, then interpret the dual prices of protein and fiber.

Source: Practice problem

Problem type	MIN, 2 variables, 3 constraints
Status	OPTIMAL
Optimal Cost (\$)	437.64706
Dual status	OPTIMAL
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Course	Sensitivity Analysis and Duality (Dr. Islam Ali)
Date	28 September 2026, 08:45



1. Model formulation

MIN $z = 0.3 \text{ Corn} + 0.9 \text{ Soy}$
 subject to
 $\text{Corn} + \text{Soy} \geq 800$ (Total)
 $0.21 \text{ Corn} - 0.3 \text{ Soy} \leq 0$ (Protein)
 $0.03 \text{ Corn} - 0.01 \text{ Soy} \geq 0$ (Fiber)
 $\text{Corn} \geq 0, \text{ Soy} \geq 0$

Data table

Constraint	Corn	Soy	Sense	RHS
Total	1	1	$\geq$	800
Protein	0.21	-0.3	$\leq$	0
Fiber	0.03	-0.01	$\geq$	0
Objective c	0.3	0.9	min	—

Definitions

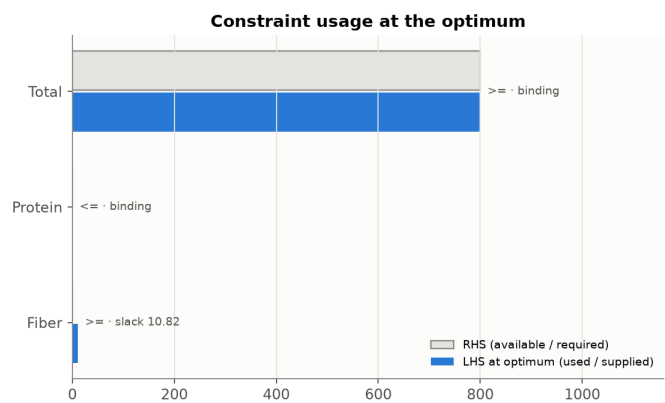
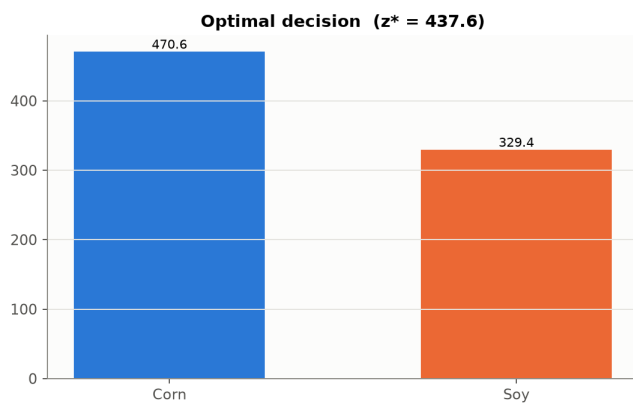
- **Corn:** kg of corn
- **Soy:** kg of soybean meal
- **Total:** kg of feed needed
- **Protein:** protein share at least 30% (rearranged)
- **Fiber:** fiber share at most 5% (rearranged)

2. Optimal solution

Optimal Cost (\$): **437.64706**

Variable	Description	Optimal value	Obj. coefficient	Contribution
Corn	kg of corn	470.6	0.3	141.2
Soy	kg of soybean meal	329.4	0.9	296.5

Constraint	LHS	Sense	RHS	Slack/Surplus	Binding
Total	800	$\geq$	800	0	yes
Protein	1.34e-14	$\leq$	0	-1.34e-14	yes
Fiber	10.82	$\geq$	0	10.82	no



Corner points

Corn	Soy	Cost (\$)	Optimal?
470.6	329.4	437.6	✓
200	600	600	

3. Sensitivity analysis

Objective-coefficient ranging (variable cells)

Variable	Value	Reduced cost	Obj. coef	Allow. +	Allow. –	Low	High
Corn	470.6	0	0.3	0.6	0.93	-0.63	0.9
Soy	329.4	0	0.9	∞	0.6	0.3	∞

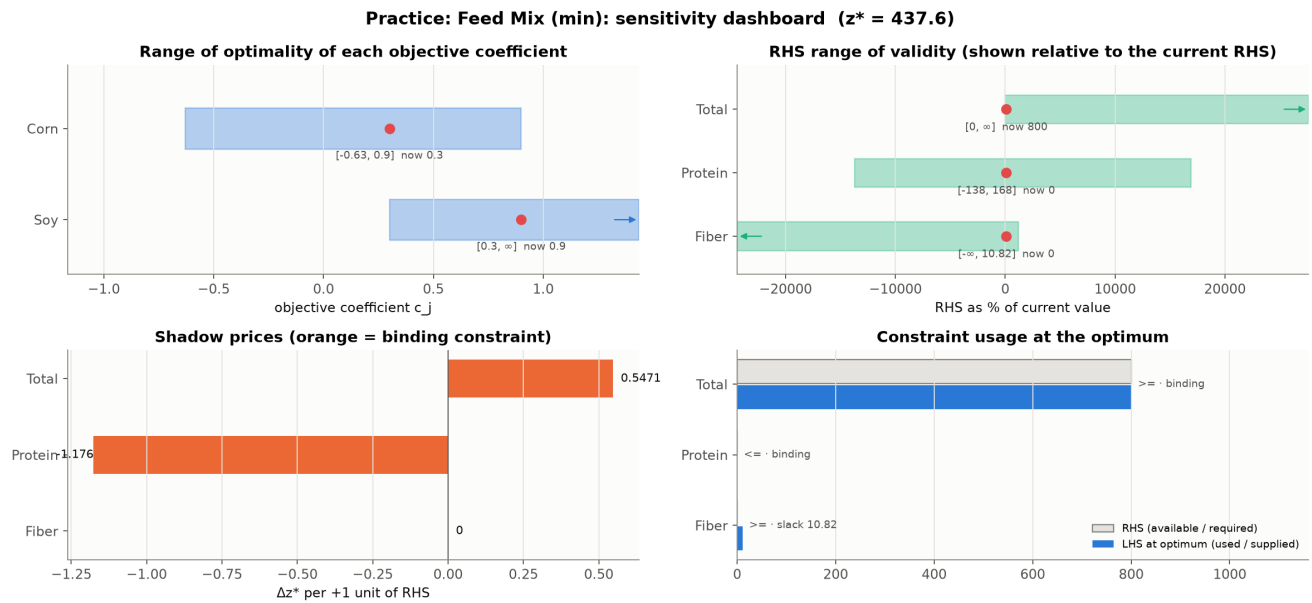
Right-hand-side ranging (constraints)

Constraint	LHS	Sense	RHS	Slack	Shadow price	Allow. +	Allow. –	Low	High
Total	800	$\geq$	800	0	0.5471	∞	800	0	∞
Protein	1.459e-15	$\leq$	0	0	-1.176	168	138	-138	168
Fiber	10.82	$\geq$	0	10.82	0	10.82	∞	$-\infty$	10.82

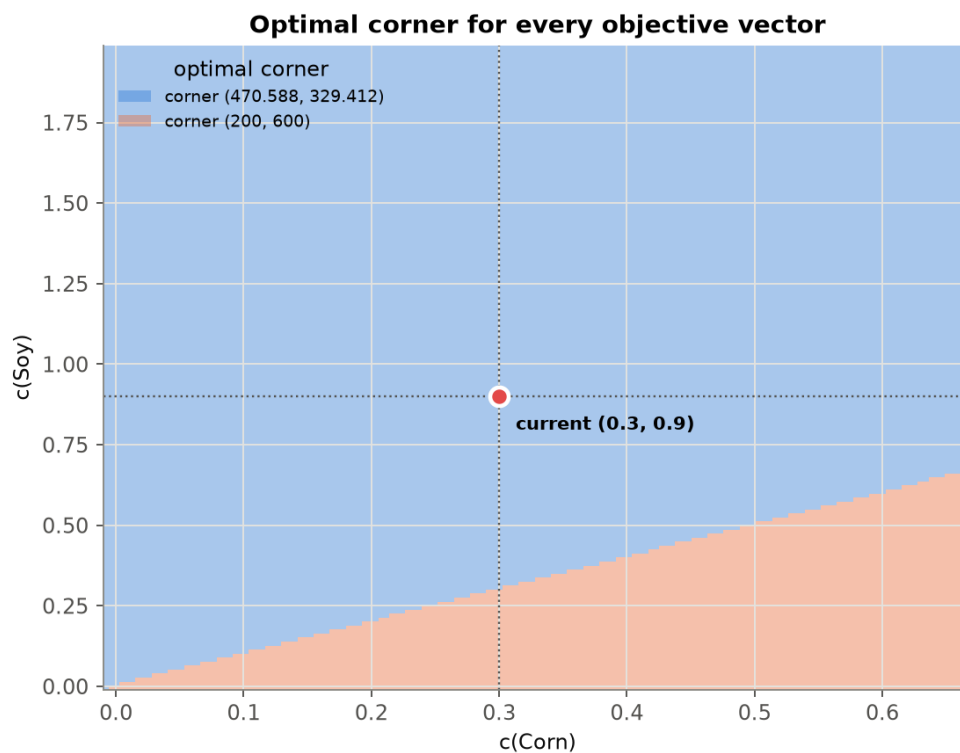
Interpretation

- **Corn** = 470.6. Its objective coefficient (0.3) may vary in $[-0.63, 0.9]$ without changing the optimal solution (one change at a time).
- **Soy** = 329.4. Its objective coefficient (0.9) may vary in $[0.3, \infty]$ without changing the optimal solution (one change at a time).
- **Total** is binding. Shadow price 0.5471: each extra unit of RHS changes Cost (\$) by 0.5471, valid while the RHS stays in $[0, \infty]$.
- **Protein** is binding. Shadow price -1.176: each extra unit of RHS changes Cost (\$) by -1.176, valid while the RHS stays in $[-138, 168]$.
- **Fiber** is not binding (slack 10.82), so its shadow price is 0; the RHS can move within $[-\infty, 10.82]$ without effect on Cost (\$).

Sensitivity dashboard

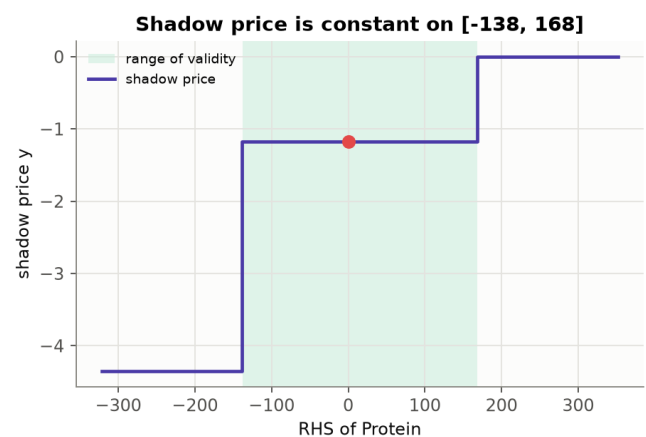
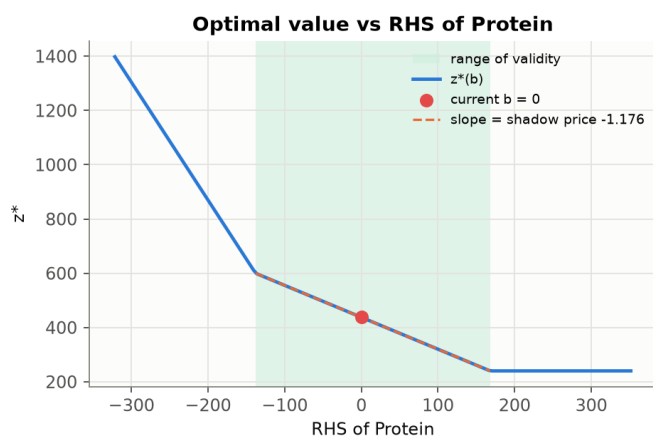
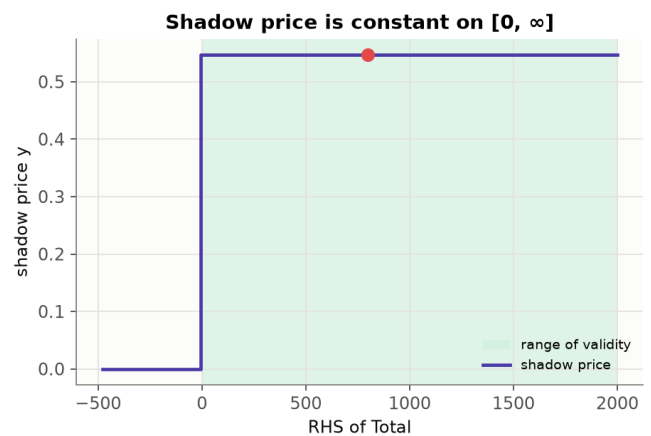
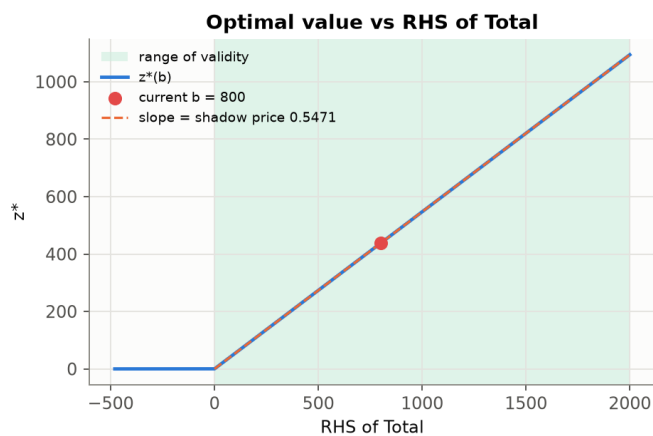
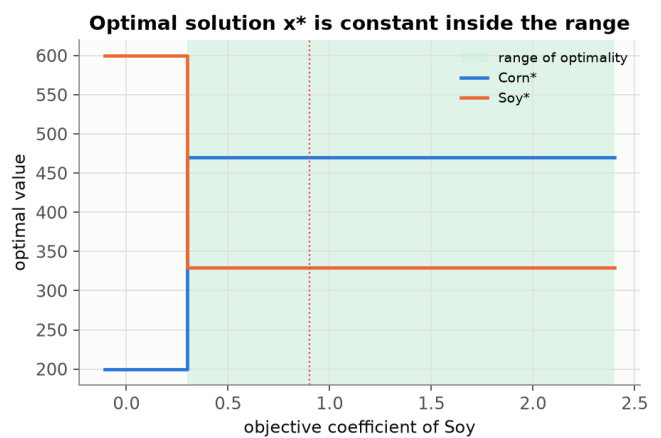
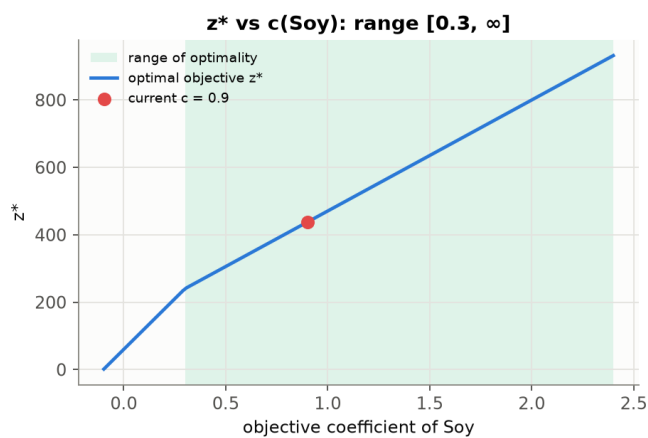
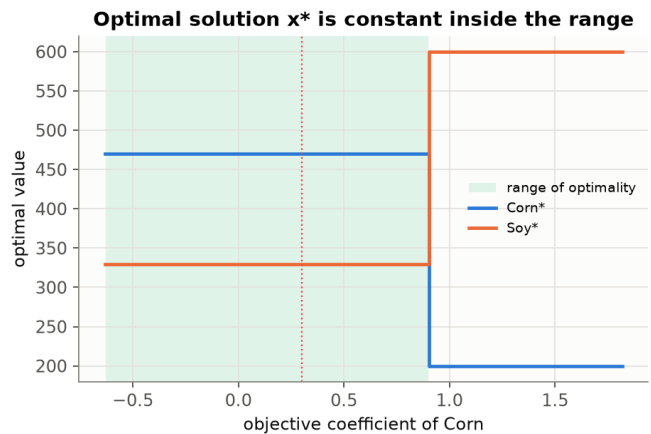
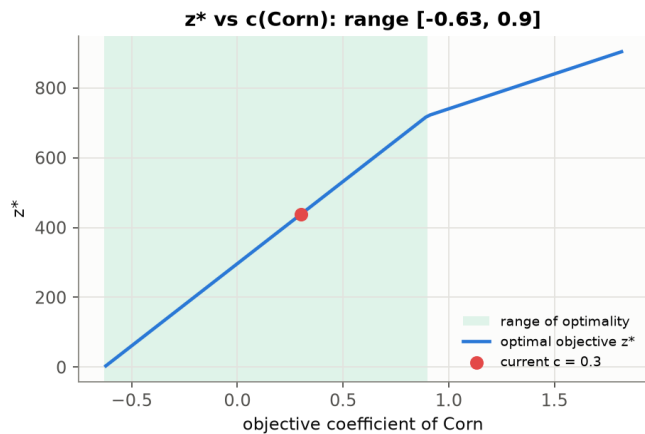


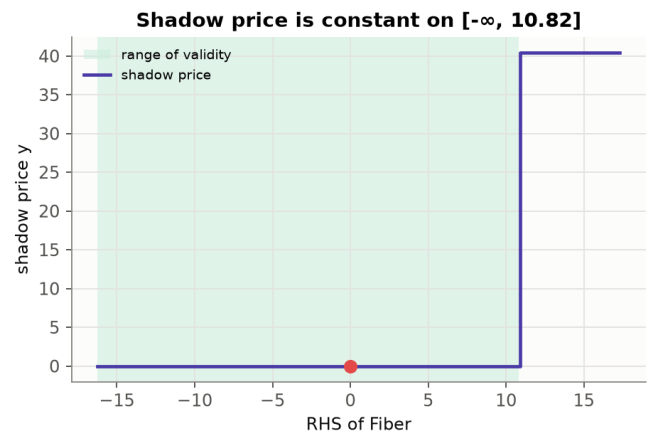
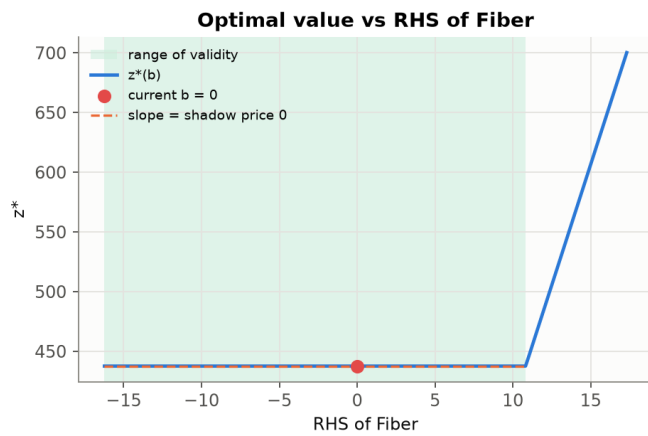
Optimal corner for every objective vector



4. Parametric analysis

Each chart re-solves the LP for a range of values of one coefficient. The shaded band is the range reported in section 3.

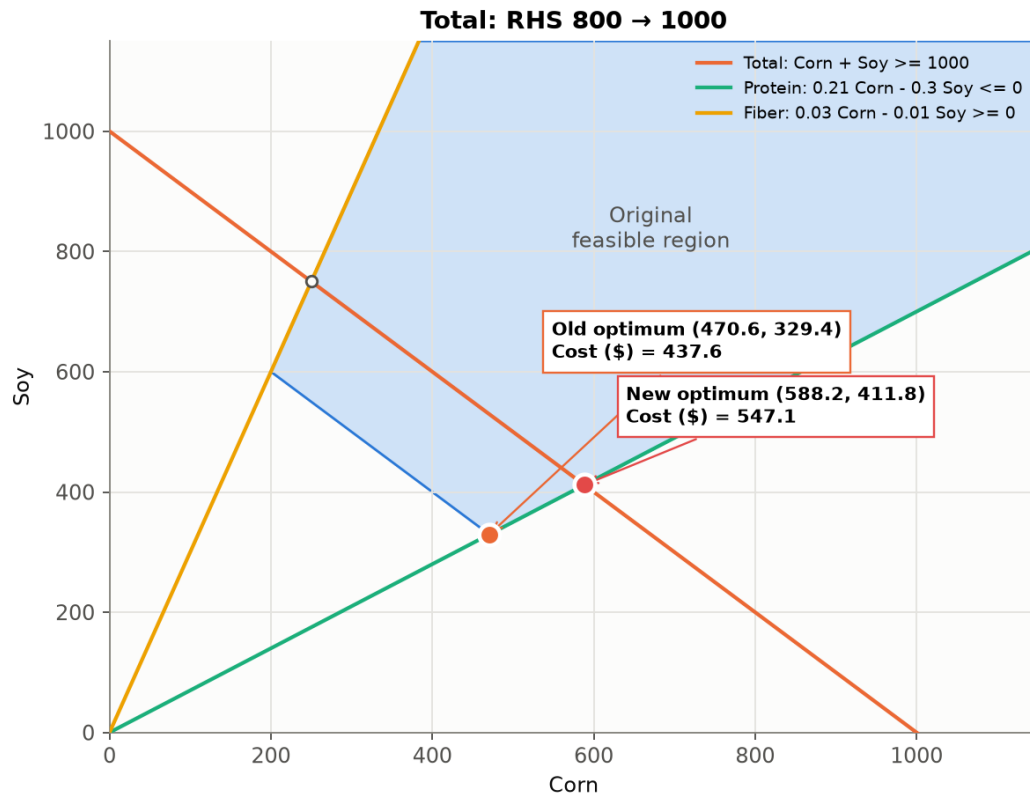




5. What-if scenarios

'Predicted' uses only the sensitivity report (shown when the change is inside its range); 'Actual' re-solves the LP.

Scenario	InRange	Predicted	Actual	NewSolution	Note
b(Total): 800 → 1000	yes	547.1	547.1	588.2, 411.8	feed requirement rises to 1000 kg



6. Duality

Dual formulation

MAX $z = 800 y_{\text{Total}}$
 subject to
 $y_{\text{Total}} + 0.21 y_{\text{Protein}} + 0.03 y_{\text{Fiber}} \leq 0.3$ (Corn)
 $y_{\text{Total}} - 0.3 y_{\text{Protein}} - 0.01 y_{\text{Fiber}} \leq 0.9$ (Soy)
 $y_{\text{Total}} \geq 0, y_{\text{Protein}} \leq 0, y_{\text{Fiber}} \geq 0$

Dual solution (resource / requirement prices)

Dual variable	Primal constraint	Value y^*	Primal shadow price	$b_i \cdot y_i$
y_{Total}	Total	0.5471	0.5471	437.6
y_{Protein}	Protein	-1.176	-1.176	-0
y_{Fiber}	Fiber	0	0	0

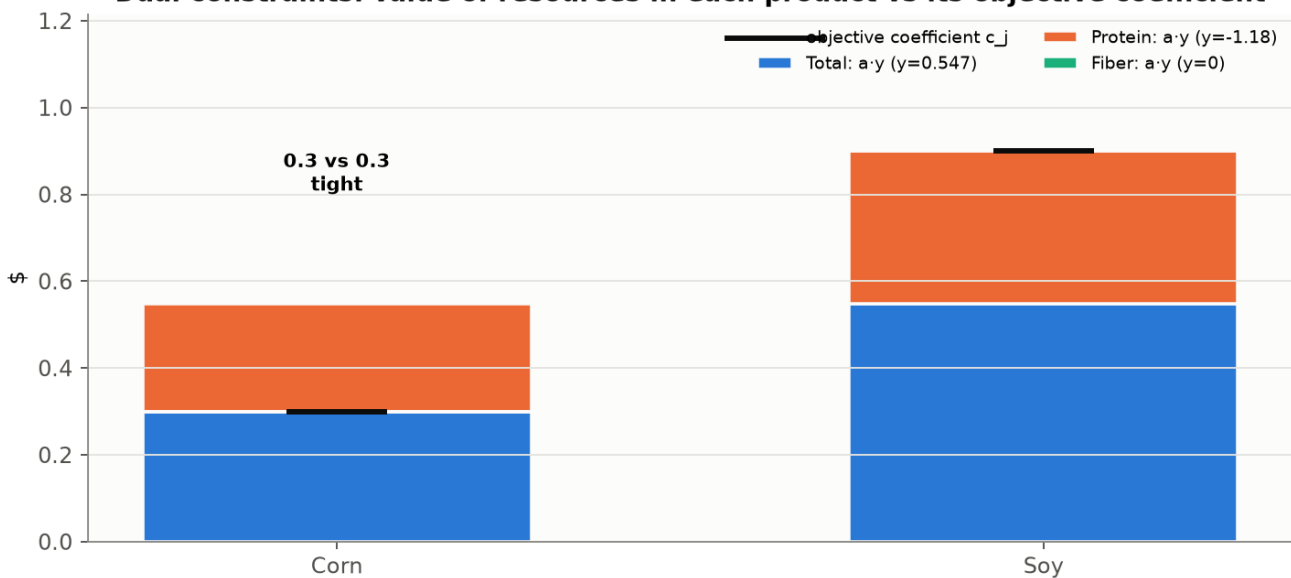
Strong duality check: primal $z^* = 437.64706$, dual $w^* = 437.64706$ (equal ✓). Dual of the dual equals the primal: yes.

Complementary slackness (all products must be 0)

Pair	Slack	DualValue	Product
Total slack $\times y_{\text{Total}}$	0	0.5471	0
Protein slack $\times y_{\text{Protein}}$	-1.34e-14	-1.176	0
Fiber slack $\times y_{\text{Fiber}}$	10.82	0	0
Corn $\times$ surplus of dual constraint Corn	-5.551e-17	470.6	0
Soy $\times$ surplus of dual constraint Soy	-1.11e-16	329.4	0

0.9 vs 0.9
tight

Dual constraints: value of resources in each product vs its objective coefficient

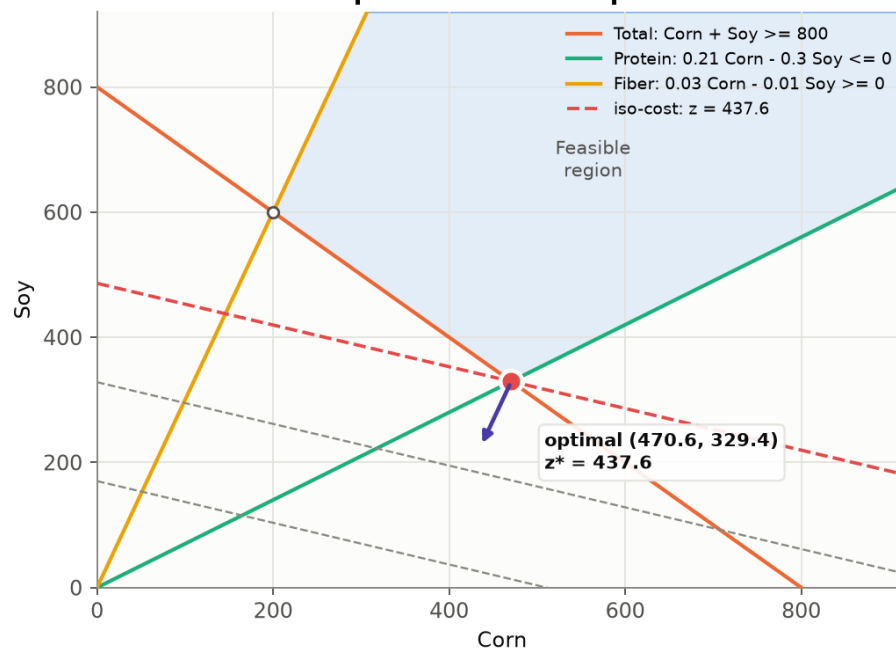


Primal (min): 3 constraints × 2 variables

0.3	0.9	← objective
1	1	≥ 800
0.21	-0.3	≤ 0
0.03	-0.01	≥ 0
Corn	Soy	

Dual (max): 2 constraints × 3 variables

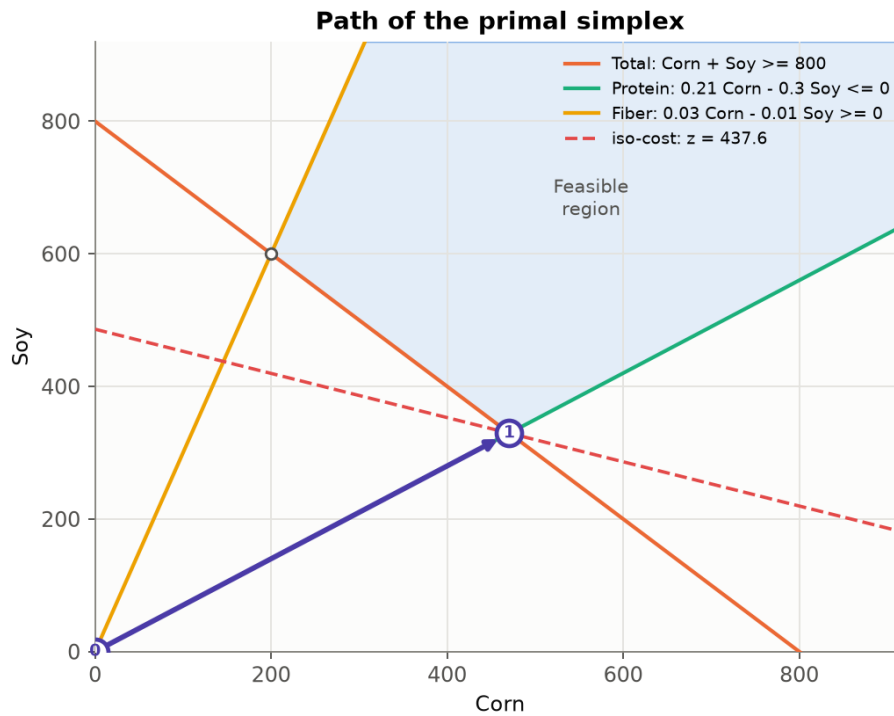
800	0	0	← objective
1	0.21	0.03	≤ 0.3
1	-0.3	-0.01	≤ 0.9
y_Total	y_Protein	y_Fiber	

Graphical view of the primal

7. Simplex iterations

Two-phase tableau simplex (entering column = most negative $z_j - c_j$; leaving row = minimum ratio).

Tableau	Method	Phase	Entering	Leaving	Objective	PrimalFeasible	DualFeasible	Point
0	primal	1	Corn	s2	800	yes	no	(0, 0)
1	primal	1	Soy	a3	800	yes	no	(0, 0)
2	primal	1	e3	a1	800	yes	no	(0, 0)
3	primal	1			-0	yes	yes	(470.6, 329.4)
4	primal	2			437.6	yes	yes	(470.6, 329.4)



Primal simplex - phase 1, tableau 0
enter Corn, leave s2 (min ratio 0)

	Corn	Soy	e1	s2	e3	a1	a3	RHS
w	-1.03	-0.99	1	0	1	0	0	-800
a1	1	1	-1	0	0	1	0	800
s2	0.21	-0.3	0	1	0	0	0	0
a3	0.03	-0.01	0	0	-1	0	1	0

Primal simplex - phase 1, tableau 1
enter Soy, leave a3 (min ratio 0)

	Corn	Soy	e1	s2	e3	a1	a3	RHS
w	0	-2.46	1	4.9	1	0	0	-800
a1	0	2.43	-1	-4.76	0	1	0	800
Corn	1	-1.43	0	4.76	0	0	0	0
a3	0	0.0329	0	-0.143	-1	0	1	0

Primal simplex - phase 1, tableau 2
enter e3, leave a1 (min ratio 10.82)

	Corn	Soy	e1	s2	e3	a1	a3	RHS
w	0	0	1	-5.8	-73.9	0	74.9	-800
a1	0	0	-1	5.8	73.9	1	-73.9	800
Corn	1	0	0	-1.45	-43.5	0	43.5	0
Soy	0	1	0	-4.35	-30.4	0	30.4	0

Primal simplex - phase 1, tableau 3
end of phase 1

	Corn	Soy	e1	s2	e3	a1	a3	RHS
w	0	0	0	0	0	1	1	0
e3	0	0	-0.0135	0.0784	1	0.0135	-1	10.8
Corn	1	0	-0.588	1.96	0	0.588	0	471
Soy	0	1	-0.412	-1.96	0	0.412	0	329

Primal simplex - phase 2, tableau 4
optimal tableau

	Corn	Soy	e1	s2	e3	RHS
z	0	0	0.547	1.18	0	-438
e3	0	0	-0.0135	0.0784	1	10.8
Corn	1	0	-0.588	1.96	0	471
Soy	0	1	-0.412	-1.96	0	329